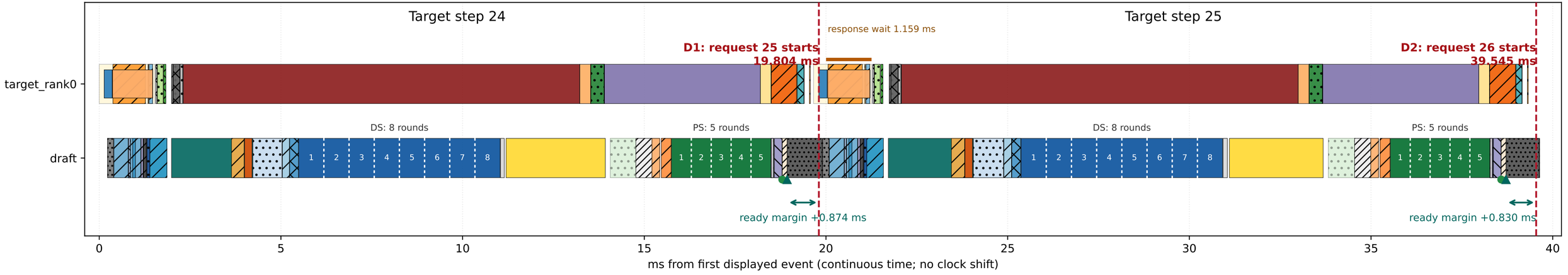
LayerSkip Llama2-7B + llama-68m | Before: selected prepack base | P1 hit -> P1 hit | UID 461_t0 | steps 24-25

Local orion RTX4090; B1 T0.7 BF16; K8/5 E22 DS4/W32 PS15/W15 C3 M15; native profile ON, 12 inputs output256; no forward GPU stamps | Before: selected prepack base | B=1, K1/K2 8/5

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	385	42.636	3.088	157.106
P2 hit	409	45.293	2.369	120.524
Miss	109	12.071	1.890	89.515
Overall	**903**	**87.929**	**2.618**	**131.998**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	385	19.657	6.975	+3.070	0.683	4.985	+0.626	0.548
P2 hit	409	19.657	7.011	+3.028	0.690	4.972	+0.648	0.548
Miss	109	21.113	6.596	+4.843	0.627	4.972	+0.794	0.548
Overall	**903**	**19.833**	**6.945**	**+3.265**	**0.679**	**4.977**	**+0.656**	**0.548**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_response_topology_prepare_p1
tree_response_topology_prepare_p2
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

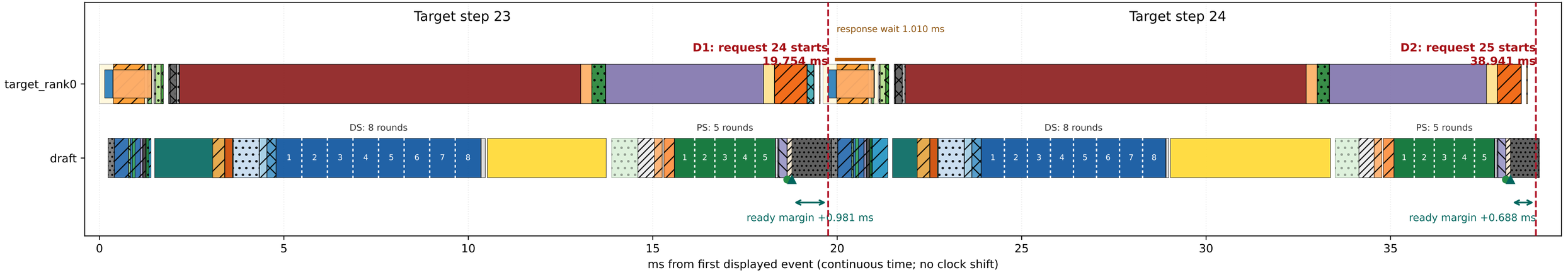
LayerSkip Llama2-7B + llama-68m | Before: selected prepack base | P2 hit -> P2 hit | UID 221_t0 | steps 23-24

Local orion RTX4090; B1 T0.7 BF16; K8/5 E22 DS4/W32 PS15/W15 C3 M15; native profile ON, 12 inputs output256; no forward GPU stamps | Before: selected prepack base | B=1, K1/K2 8/5

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	385	42.636	3.088	157.106
P2 hit	409	45.293	2.369	120.524
Miss	109	12.071	1.890	89.515
Overall	**903**	**87.929**	**2.618**	**131.998**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	385	19.657	6.975	+3.070	0.683	4.985	+0.626	0.548
P2 hit	409	19.657	7.011	+3.028	0.690	4.972	+0.648	0.548
Miss	109	21.113	6.596	+4.843	0.627	4.972	+0.794	0.548
Overall	**903**	**19.833**	**6.945**	**+3.265**	**0.679**	**4.977**	**+0.656**	**0.548**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_response_topology_prepare_p1
tree_response_topology_prepare_p2
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

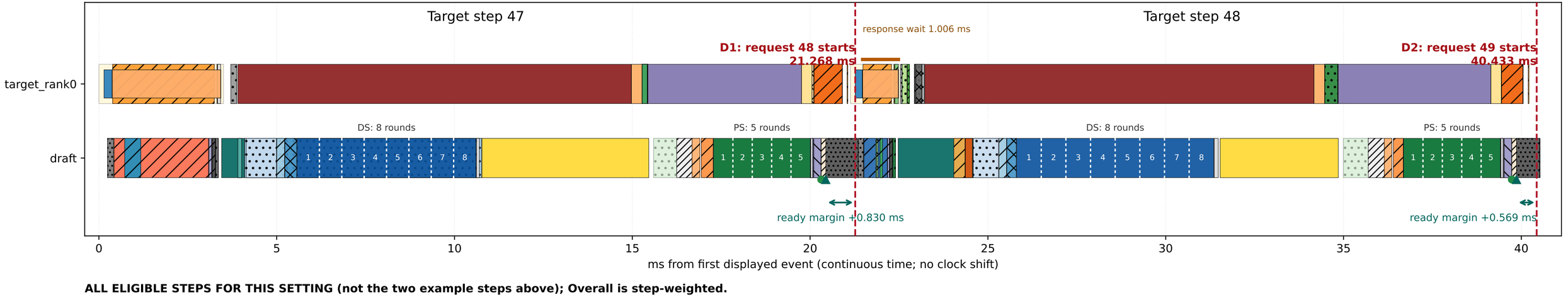
LayerSkip Llama2-7B + llama-68m | Before: selected prepack base | Miss -> P2 hit | UID 221_t0 | steps 47-48

Local orion RTX4090; B1 T0.7 BF16; K8/5 E22 DS4/W32 PS15/W15 C3 M15; native profile ON, 12 inputs output256; no forward GPU stamps | Before: selected prepack base | B=1, K1/K2 8/5

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

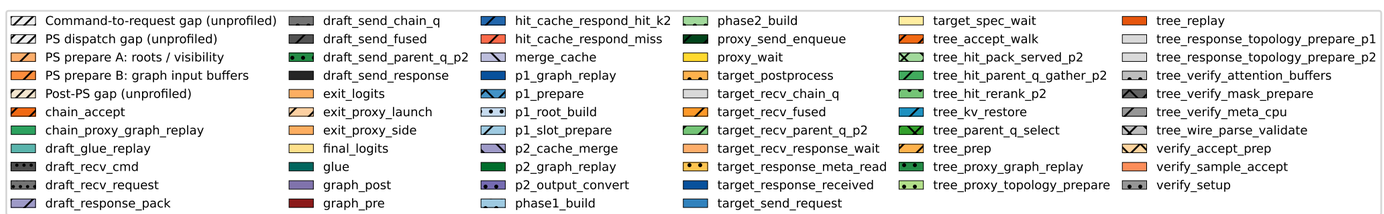
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	385	42.636	3.088	157.106
P2 hit	409	45.293	2.369	120.524
Miss	109	12.071	1.890	89.515
Overall	**903**	**87.929**	**2.618**	**131.998**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	385	19.657	6.975	+3.070	0.683	4.985	+0.626	0.548
P2 hit	409	19.657	7.011	+3.028	0.690	4.972	+0.648	0.548
Miss	109	21.113	6.596	+4.843	0.627	4.972	+0.794	0.548
Overall	**903**	**19.833**	**6.945**	**+3.265**	**0.679**	**4.977**	**+0.656**	**0.548**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

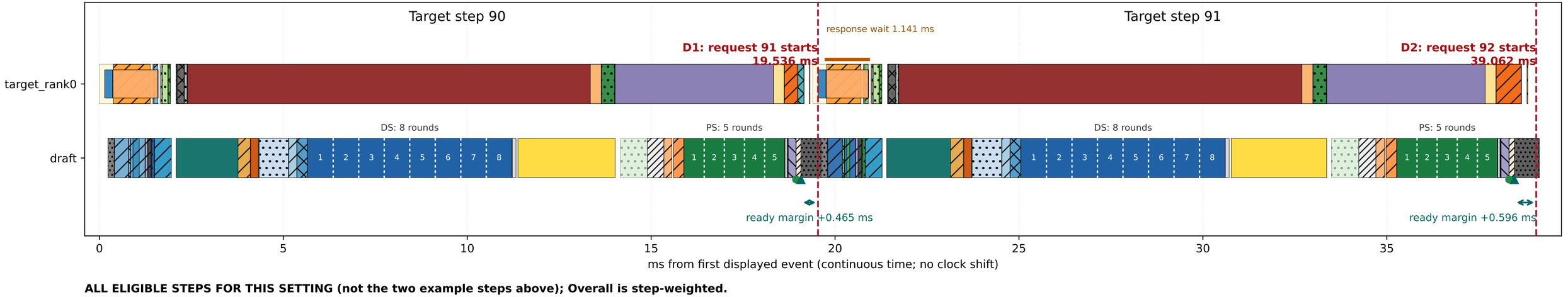
LayerSkip Llama2-7B + llama-68m | After: redundant prep removed | P1 hit -> P2 hit | UID 181_t0 | steps 90-91

Local orion RTX4090; B1 T0.7 BF16; K8/5 E22 DS4/W32 PS15/W15 C3 M15; native profile ON, 12 inputs output256; no forward GPU stamps | After: redundant prep removed | B=1, K1/K2 8/5

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	385	42.636	3.088	159.158
P2 hit	409	45.293	2.369	122.075
Miss	109	12.071	1.890	90.941
Overall	**903**	**87.929**	**2.618**	**133.759**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	385	19.404	6.851	+3.218	0.683	4.843	+0.683	0.548
P2 hit	409	19.408	6.891	+3.177	0.689	4.861	+0.679	0.547
Miss	109	20.782	6.454	+4.959	0.626	4.835	+0.828	0.547
Overall	**903**	**19.572**	**6.821**	**+3.410**	**0.679**	**4.850**	**+0.699**	**0.548**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_response_topology_prepare_p1
tree_response_topology_prepare_p2
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

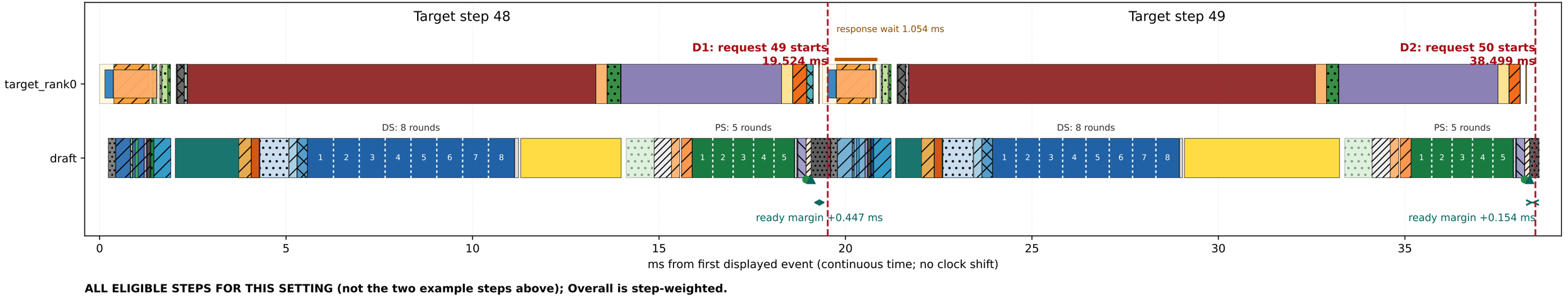
LayerSkip Llama2-7B + llama-68m | After: redundant prep removed | P2 hit -> P1 hit | UID 101_t0 | steps 48-49

Local orion RTX4090; B1 T0.7 BF16; K8/5 E22 DS4/W32 PS15/W15 C3 M15; native profile ON, 12 inputs output256; no forward GPU stamps | After: redundant prep removed | B=1, K1/K2 8/5

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	385	42.636	3.088	159.158
P2 hit	409	45.293	2.369	122.075
Miss	109	12.071	1.890	90.941
Overall	**903**	**87.929**	**2.618**	**133.759**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	385	19.404	6.851	+3.218	0.683	4.843	+0.683	0.548
P2 hit	409	19.408	6.891	+3.177	0.689	4.861	+0.679	0.547
Miss	109	20.782	6.454	+4.959	0.626	4.835	+0.828	0.547
Overall	**903**	**19.572**	**6.821**	**+3.410**	**0.679**	**4.850**	**+0.699**	**0.548**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_response_topology_prepare_p1
tree_response_topology_prepare_p2
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

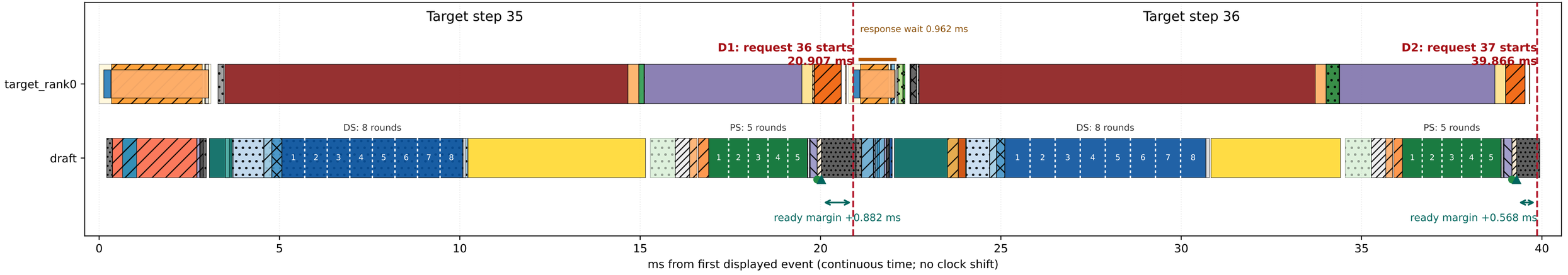
LayerSkip Llama2-7B + llama-68m | After: redundant prep removed | Miss -> P1 hit | UID 461_t0 | steps 35-36

Local orion RTX4090; B1 T0.7 BF16; K8/5 E22 DS4/W32 PS15/W15 C3 M15; native profile ON, 12 inputs output256; no forward GPU stamps | After: redundant prep removed | B=1, K1/K2 8/5

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
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Miss	109	12.071	1.890	90.941
Overall	**903**	**87.929**	**2.618**	**133.759**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	385	19.404	6.851	+3.218	0.683	4.843	+0.683	0.548
P2 hit	409	19.408	6.891	+3.177	0.689	4.861	+0.679	0.547
Miss	109	20.782	6.454	+4.959	0.626	4.835	+0.828	0.547
Overall	**903**	**19.572**	**6.821**	**+3.410**	**0.679**	**4.850**	**+0.699**	**0.548**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_response_topology_prepare_p1
tree_response_topology_prepare_p2
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.